ANNEX 1 –Reporting

Instructions

This is a template for the CanSat report for the **Preliminary Design Review, Critical Design Review and Final Design Review**– please copy and paste the below template into a separate document and follow it as it is. **Please do not include any personal information such as names or photographs of team members or anybody involved in outreach activities in the reports.**

Preliminary Design Review (PDR)

The first report should include your plans for the project and initial activities. Include high level descriptions, diagrams and basic sketches. Consider what software and hardware could be used to achieve both missions and how feasible they will be to use. Include calculations and identify suitable designs. Show your plans for testing.

Critical Design Review (CDR)

The CDR should include evidence of prototypes through photos or CAD sketches. Think about including high level systems diagrams, circuit diagrams, block diagrams etc. Explain what sensors are being used and the justification for using them. Include block and flow diagrams and a high-level description of the design as the software is starting to be put together and tested. A full copy of the code is not required. Include parachute design, calculations used to reach the final decision and information on any prototypes developed.

Important: the CDR must show the results of the testing performed and details of the decisions made based on the results; the report must also show evidence that the CanSat is functional and include at least one photograph of the CanSat itself (not just the parachute).

Final Design Review (FDR)

The final design review is submitted by finalists only. The third report should cover the final iteration of the mechanical design, electronics, software, landing and recovery system and ground support system of the CanSat. If changes were made include why. Design justifications are important for each sketch or CAD rendering etc. Include all finalised tests and decisions made based on them. Results could be displayed in a table of tests and results, and there must be verification against the CanSat requirements (found on page 8 of the competition guidelines).

For the FDR we have included sections on the launch day and lessons learned to be included. Teams invited to the final should also start to prepare a 7-8 minute presentation that will include aspects of this report but also a placeholder for final results from the final launch day – this will be presented on the day after the launch.

Page limits

Maximum page limits for the reports are:

Preliminary Design Review – 20 pages (plus unlimited appendices)
Critical Design Review - 25 pages (plus unlimited appendices)
Final Design Review - 30 pages (plus unlimited appendices)

Any pages submitted over these maximum amounts will not be read.

UK CanSat Competition

Project Pathfinder

School: Trinity Croydon School

Preliminary Design Review

Critical Design Review

Final Design Review

Date: -------

Introduction

1.1 Team Organisation and Roles

This section should contain a simple list of the assigned roles and responsibilities. ***It must be anonymous and not include any personal data such as names or photographs. Instead of writing a name/nickname, write 'Student 1', 'Student 2' etc, or simply list the roles with no reference to the student.***

Student 1 – Project Manager, Launch day Manager, Programmer, Tester, Team Leader

Student 2 – Mechanical Engineer, Launch day Manager, Outreach, Tester, CAD Lead

Student 3 – Calculations, Planning Manager, Tester, Mechanical Engineer

Student 4 – Electrical Engineer, Tester

Student 5 – Testing Manager, Parachute Architect

Student 6 – Outreach Lead

Project manager: Coordinates tasks between different people working on the project

Programmer: Tasked with writing the programs for the electrical equipment

Tester: Involved in testing different components of the CANSAT

Outreach: Member of the outreach team whose focus is to make our project known to the masses

Mechanical engineer: Member of the team responsible for constructing the CANSAT

Calculations: Performs necessary calculations for different components of the CANSAT

Planning Manager: Ensures the team is well informed of the key dates for the CANSAT project

Electrical engineer: Member of the team who manages the electrical circuitry of the CANSAT

CAD lead: Directs the team which is in charge of producing a CAD design for the CANSAT

Testing manager: Coordinates the team who is responsible for testing components of the CANSAT

Parachute architect: Designs and manufactures the parachute

Outreach lead: In charge of the team that ensures the project is known to the masses

1.2 Mission Overview

1.2.1 Mission Objectives

Primary Mission:

For our primary mission, we will be measuring the air temperature and air pressure every second and transmitting these results back to the ground. This is where they can be analysed and documented to use later on once all results have been compiled as per the given requirements of the CanSat mission. For the temperature and air pressure, a BME280 sensor can be used then the data can be stored on an SD card. We are storing this information on the SD card, as the transmission on the day could be faulty, fail entirely, or there could be any sort of errors within the system. The SD card, therefore, acts as a form of redundancy/ backup. This also allows us to have data which, can be indexed and can provide the specific temperature and pressure for any given altitude or time. This would be useful in a real satellite, to find out the composition of a planetary bodies' atmosphere.

Commented [JM1]: Remember the primary mission as outlined in the brief is to transmit the temperature and pressure data. It would be good to state why you are also storing the data to an SD card. You could mention that this is a form of redundancy/back up to make sure data is backed up in case data transmission fails. This is an important aspect of designing a satellite.

We are using the BME280 as it measures temperature, air pressure and humidity. This means that we can record three different quantities with one device, reducing the number of components we need. Its dimensions are 18x11.5x3mm, which is an added benefit, as it is small and compact, leaving space for other potential components. The BME280 has the following inaccuracies:

- Temperature: Accuracy of around ±1°C.
- Pressure: Accuracy of around ±1 hPa.
- Humidity: Accuracy of around ±3%.

These inaccuracies are negligible for the data we are trying to collect, therefore the BMP280 is the best choice for our primary mission due to its space efficiency and its accuracy in measuring temperature and air pressure (and humidity, however this is not part of the brief).

We are also using the Arduino Nano-ESP32, as it is relatively smaller than other options, and has sufficient connections for the many sensors we plan to use. The Arduino Nano-ESP32 is also programmed using C++, which is a just what we need, as it provides low-level control over each sensor and the individual pin connections. It being a low-level programming language also means it is highly efficient as there is no unnecessary code which accompanies higher level languages.

Commented [JM2]: Are there other benefits to the Arduino nano you could mention here, such as the language you use to program it, the range of input/ outputs it has etc.

Secondary Mission:

For our secondary mission we will use a range of 2D images taken, by a the OV6542 camera module, from a number of angles to create a topographical map of the landscape, as the different angles will allow us to determine the depth coordinate. The various angles will be provided by the automatic drift the cansat will have. All the images that are taken will be stored on an SD drive, as active transmission would be slow, and more importantly ineffective, as to create the topographical map, we require the use of PixPro. A software that will convert a series of our 2D images into a topographical map, given that we provide the Altitude, Longitude, and Latitude of each photo. For this we have a GY-NEo6MV2 GPS, which will record the required data for each picture, and store it in a dictionary, next to the photo. PixPro will be installed on a ground-based computer, and the topological map will be processed after the satellite has been recovered and the photos have been downloaded off the SD drive. Of course whilst this camera has the capability of capturing in 5 megapixels, which is good for a standard camera, there are alternatives which may be better at capturing the landscape from the 400 metre altitude which the CanSat will be located. Many applications for photogrammetry, including examples like the Orion12MP, recommend at least 12 megapixels; higher resolutions such as 24 or 36 megapixels are also used, as they are better suited for more detailed capturing. However, we are limited for choosing higher resolution cameras due to the greater required storage space and processing power. Our equipment would not be able to maintain the required processing power for these cameras, and the SD card only has 16 Gigabytes of storage space, not enough for the number of images we need to capture for the 3D render. Thus, a camera with more megapixels is not compatible for our CanSat. Furthermore, few cameras are compatible with our Arduino nano, so finding a new camera with higher resolution and compatibility with the arduino nano is not reliable, so the OV6542 is our best option.

Our secondary (mission of active stabilisation) will assist with ensuring our topographical mapping is accurate and precise by adjusting the angle of the CanSat to take the required pictures, and by stabilising it when there is wind, or the parachute is unstable. However, as it is only a reaction to some of the forces acting on it, is has no control over where it goes. This could be implemented so that the cansat follows a large circular path to get all the angels of a certain area, however this would make the active stabilization much harder to do with the same components, as they would be both attempting to follow the path, and keep the CanSat steady at the same time.

Commented [JM3]: Does the system used to stabilise the satellite also used to position it in these different angles needed for the 3D rendering? It would be good to comment on how this will be done and perhaps how many different angles you will need to make the 3D rendering.

The reason behind this being our secondary mission is because on (planets) outside our solar system, the satellite may be adventuring into uncharted territory, so the terrain will be unpredictable, therefore we can use the topographical map as information to (learn about the new planet). From a topological map, we can:

Commented [JM4]: Make sure to say that it is likely to be a planet outside our solar system. As we would be able to observe many of the planets in our solar system using telescopes, so we wouldn't need to use a satellite.

Commented [JM5]: Make sure to mention what things you want to learn about the planet, like habitability, ability to grow food, obtain clean drinking water, or existence of raw materials like minerals (can you obtain this from just photographs alone?)

- Learn about the geological history of the planet:
  - By analysing mountains, valleys, craters, and volcanoes which can also provide information about the planet's interior processes.
  - By analysing the general shape of the terrain, to infer how it has changed over time, such as through tectonic activity, meteorite impacts, etc.…
- Assist with the primary mission, as elevation differences can influence wind patterns, and temperature fluctuations which are essential in understanding a planet's atmosphere and weather.
- Locate possible landing sites for potential astronauts.
  - Depending on their assigned mission, astronauts may require a high or low altitude, which can be easily identified by a topographical map.
- Could allow scientist to form trends or similarities with that of earth, therefore allowing them to learn more about the new planet or moon.

The stabilisation will also allow the CanSat to land safely without damage to ensure all the Primary mission readings can be taken successfully.

The photos will need to be stable, and to assist this the active stabilisation will ensure the photos are relatively clear, and focused by reducing swinging, and turning via roll and pitch. This will be measured by the MPU-6050 Gyroscope which relays the data to the Arduino Nano-ESP32 which then corrects for the change in roll or pitch. This can be done in many ways, such as using propellers, fins, or using a counterweight that moves within the Cansat. If propellers are used, it will require more intricate code, that precisely counteracts any micro movements made by the cansat.

1.2.2 What will you measure, why and how?

Air pressure:

Knowing the value of the air pressure is important for indicating the current weather and weather to come. For example, an increase in air pressure indicates that the temperature must increase, and the opposite effect will occur when the air pressure decreases. We can compare these results to data recorded over the past few decades and assess the extent of which climate change has impacted the air environment. This will be measured using a BMP280 and then stored on an SD card.

Air temperature:

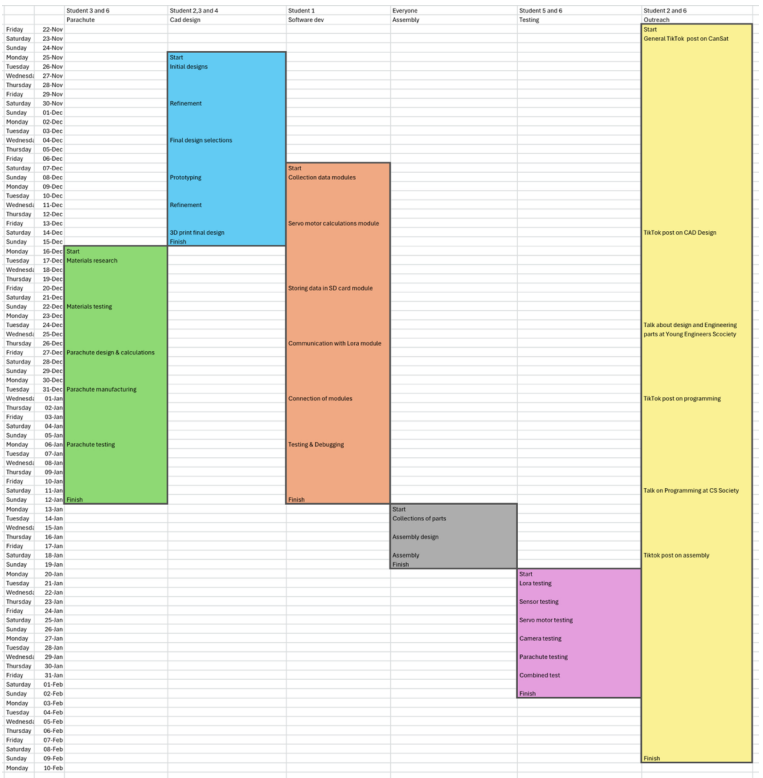
Knowing the value of the air temperature is important for similar reasons as to that of the air pressure. As it can also be used to evaluate the exacerbation of climate change and its impact on the environment.

Secondary topographical map:

We can use prixpro to transform the 2D images that we obtain from camera in the CanSat into a topographical map which will give us an accurate view of the landscape which we launched the CanSat around. This will assist with overall secondary mission of exploring distant planets. As the camera is only 5 megapixels, the quality of the picture won't be at a high enough degree of accuracy to produce a model that can be used in the real world. However, there are many limitations on the camera we can use, as it has to be small enough to fit in the cansat, it has to be simple enough for us to code it and for the images to be small enough for us to store. Therefore, the camera we have is capable for what we want to do, but won't be near enough for a real life satellite.

2 Project Planning

2.1 Time schedule



This project plan is imperative to our team's success, it provides a clear timeline that organizes tasks by role, ensuring efficient collaboration amongst all members. By breaking the project into distinct phases, such as design, development, testing, and outreach, it enables us to prioritize tasks effectively while maintaining focus on key deliverables. The chart also highlights important milestones and deadlines, promoting accountability and keeping the team aligned with project goals.

2.2 Team and External Support

We seeked help of our advising teacher, and multiple other teachers from the subjects, maths, physics, and computer science, who all helped with their respective fields relating to our CanSat.

Our advising teacher assisted us with the parachute design by providing us with a website that provides panel templates, for certain weights, and desired descent velocities. He also assisted us with the launch day risk analysis, as he had prior experience and knowledge on the matter.

The maths teacher gave us the most help by giving some useful tips on the active stabilisation, as the maths behind how much to open the flaps was quite complex.

	Project Management	Outreach	Programming	Electronics
Student 1	This student is the "group leader", he sets work that needs to be done and ensures everyone does their part. He keeps track of deadlines and is always in contact with every member of the team		This student has a lot of experience in programming, as he takes Computer Science A-level. He has also done many Coding Projects in the past which have required him to learn a basic understanding of many programming languages.	This student has also done the physical electrical work for the cansat, as he has the most familiarity with C++ and the electrical components, we are likely to use.
Student 2	This student is good at delegating work, ensuring the others are on task during work sessions, and ensuring all students have their ideas heard. He also has a strict work ethic and is enthusiastic about the cansat project.	This student is analytical in presentation planning, which is necessary when leading talks to present.		This student is also an aspiring electronics engineer, so has a lot of experience with electronics due to his past projects.
Student 3		This student has the skill set for a variety of tasks, in this project, they are working in the outreach programme, they are outgoing, and this bodes well for such a position. It is needed in order to deliver successful talks and presentations to make our project known. Additionally, this student is well connected with other members of other years, this extra networking is beneficial for the outreach programme.		
Student 4			This student has a lot of experience in working with technological programmes, performing lots of practise during their GCSE DT course. This skill is specifically strong when it comes to CAD design, where this student can form intricate designs within the programme. This can help the project with illustrating any brainstorming and ideas into a visible topographical map of the CanSat satellite.	This student has experience in working with electrical circuitry, with a defined development especially in their physics GCSE and (thus far) A-Level courses. This student can effectively and efficiently design a circuit that will meet all of the requirements and needs, whilst overcoming the restrictions of the minimal volume which the CanSat has.
Student 5		This student has experience in many areas, however, due to them not taking the computer science subject for GCSE, they have decided to focus their skill set in the outreach area. By having another person focusing on outreach, this will help to increase the publicity of the project.		
Student 6		This student has a lot of experience in many different areas but has decided to focus outreach. This student is outgoing, inquisitive and good with technology. This is very good as this student will be doing talks to mass groups of people and designing flyers to get the project out there.		

Skills  No skills 

Looking at our skills and weaknesses, we can see that the team is well rounded and covers the major areas but there are still some areas where we need to be seeking support. Programming and electronics is a very important area and we may need help when designing the CanSat. We can use our teachers to help out as they are more experienced than us. We also have access to online information which can be used, especially with programming as this is very prevalent online.

2.3 Risk Analysis

Health Risk –

Safety Risk -

Organisational Risk -

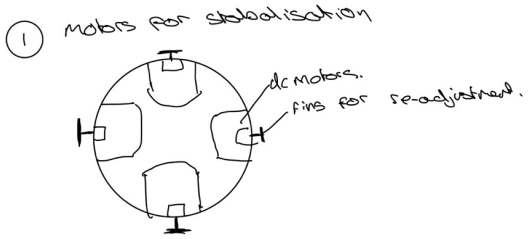
Risk	Prevention strategies
• If all the components do not fit into the case	• We will have to reduce the thickness of the wall of the casing, increase the density/ infill of the wall so it retains its structural integrity, and increase the size of the parachute so it falls at a lower rate which will decrease the likelihood that the casing is damaged on impact. • If this is insufficient, we would have to choose a component or replace with a smaller version, such as using the lowest voltage battey possible.
• Getting burns or causing fires when using soldering irons	• Anyone using the soldering irons will have training from the lead teacher before use and will always be supervised when using them.
• Use of DT equipment when constructing the CanSat including drills, saws, 3D printers and mechanical sanders. To be updated if any other equipment added to this list.	• Consult DT staff for training on any necessary equipment needed for the construction of the CanSat.
• Fire and electrocution hazard with Lithium-Ion battery used to power the CanSat.	• Anyone using or testing the CanSat with a lithium-ion battery will not use it around any liquids and will be stored safely when not being used. • The battery will also only be used in accordance with the manufacturer's specifications.
• Part failure, e.g. casing, electronics or parachute.	• For parts that we can, e.g. the casing, have a spare printed and bring it with us to the launch day. For parts such as the electronics or the parachute, we will take a repair kit so that any minor faults can be fixed before the launch. However, if a specific component fails, we have to have caught it in prior testing, where we can troubleshoot the issue and fix it, which could mean buying a replacement part, or fixing/ adapting the code.
• Risk of fall when testing the parachute or dropping the CanSat from height to test structural integrity	• We will always be supervised by an adult when performing tests from a height and we will only drop from an area where there is equipment to prevent falls, such as a handrail. • Students on the ground will wear goggles in case fragments of plastic or other materials shatter from the casing.
• Not meeting a deadline	• We will do progress checks each week to see if we are on track to meeting our deadline or no. if it becomes apparent that we may miss a deadline then we may allocate more people to a task or ask for assistance from a staff member or a relevant department.
• Active stabilisation system doesn't function properly	• We will have a section of code that activates after a certain altitude that deactivates or activates all parts of the mechanical stabilisation. This will allow us to continue taking pictures even if its is at a lower altitude.
• Battery catching fire	• We will use a battery case to prevent the lithium-ion battery from coming into contact with water or any other substances. We will also make sure that the battery is not directly beside or close to anything which can cause combustion. We will also make sure that the casing of the rocket is tight to make sure no water is able to get inside the rocket.

• Testing the parachute	• We will test the parachute with other objects of similar weight and dimensions to see if it works at different heights. We will also run different stress tests on the parachute to see if it can handle the conditions. So, we will test it when it is raining and when it is really windy to see how it fairs. If it does, then we will test the parachute with the actual cansat.

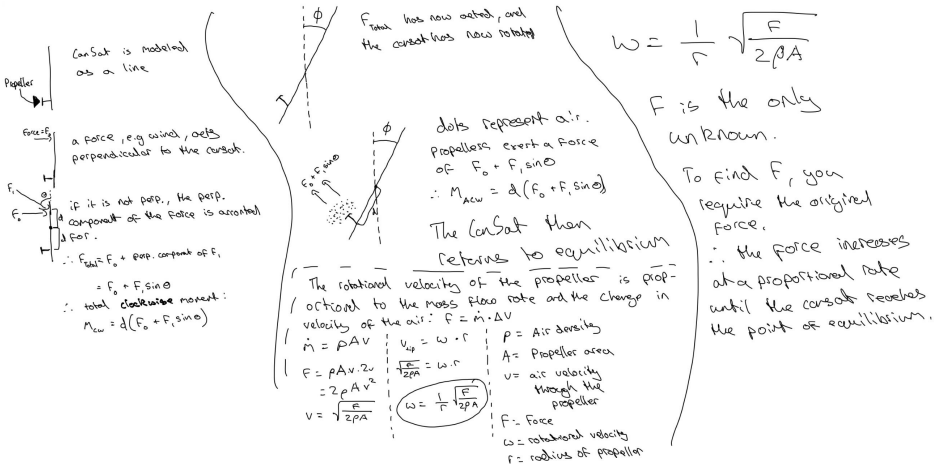
3 Cansat Design

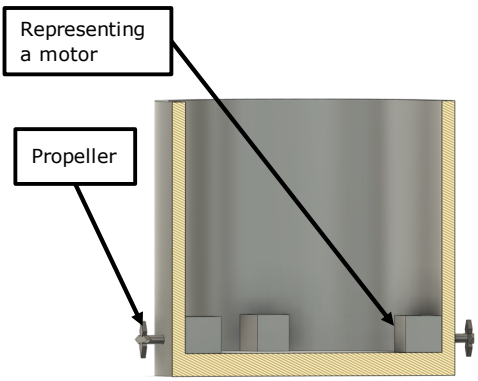
3.1 Mechanical design

Initial Ideas:

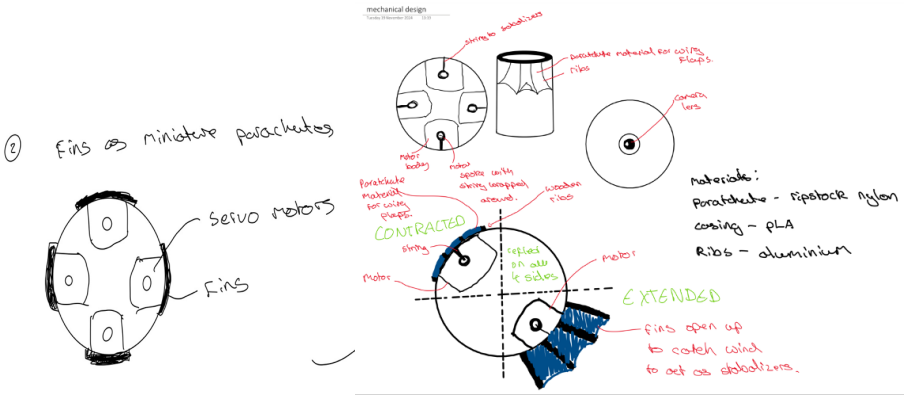


How the propellers could theoretically counteract wind.



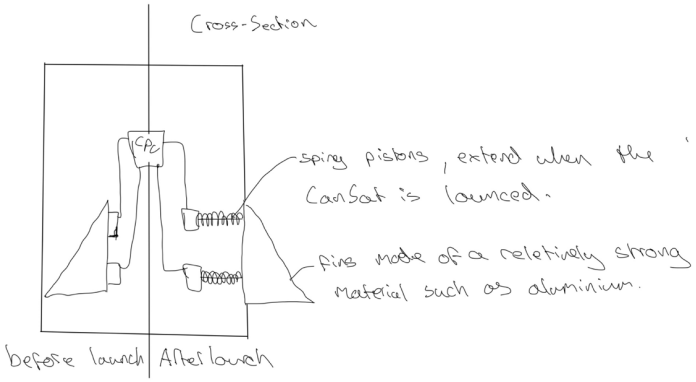


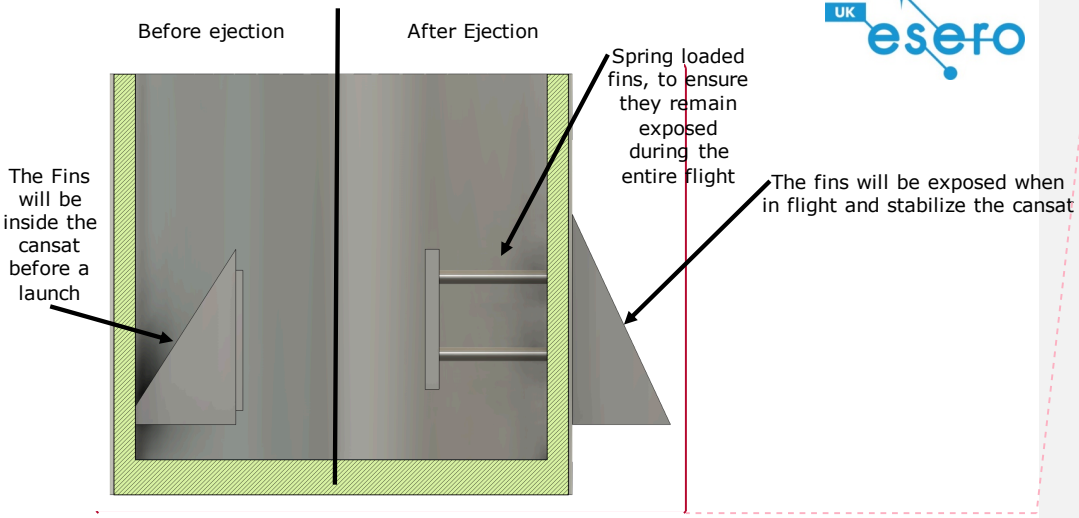
This is the first idea we had, where we would use propellers to counteract the movement caused by the wind. For example, if the Cansat faced a rotational force causing it to turn a positive(right) 23 degrees in the pitch axis, the motor on the same side would start spinning the propeller to counteract the rotation by pushing air away, which causes an equal and opposite force into the cansat causing the cansat to return to the equilibrium position. For this design, regular DC motors should be sufficient, as it has no requirement for rotation to specific angels. The propellers would be made of PLA, due to the fact that if they are light, it requires less energy to spin them at a high rate, which is what is required to 'push' enough air in the opposite direction to the rotation.



For our second idea, we had flat fins against the surface of the CanSat and can extend to catch wind to decrease the rate of descent on one of the sides for stabilization. The fins would be made of Ripstock Nylon, as it is the material best suited to the parachute, and these fins will be actin in the same way as parachutes. Ripstock nylon is also relatively thin, so it would be able to lie flush against the cansat during a launch and stay flush when not in need.

Commented [JM6]: Again refer to material, would you need something flexible so it can stay flush against the casing?

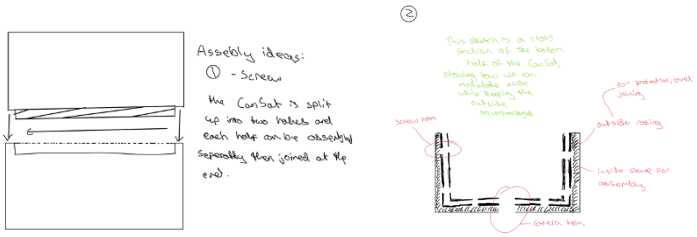




Commented [JM7]: This diagram needs annotations

For our third idea, we explored the use of fins, which would passively stabilise the cansat when ejected.

Out of the 3 initial ideas we have, we have chosen to go with the second one as it provides the best challenge, and is a solution that can actively respond to large changes in roll and pitch, whereas the first one cannot be done due to the limitations of the CanSat's size, for the propellers would protrude out of the allowed dimensions. The fin design would also be unstable at the joint and would be unable to react to large and sudden changes. They also take up too much space within the cansat, which would then become empty space after they are exposed. This is not good, as the components which have sensitive contact points and connections would have space to rattle around and potentially break. The flap design acts in the same way as the propellers, except it can lie flush with the cansat, therefore complying with the rules.



Out of these two ideas for how to separate the casing, we have chosen the first, as it is much simpler, and does not require intricate detailing of where our parts will be going in the CanSat. It also allows us to access the top half to ensure the servos and mechanical parts are all stable, and to access all the components such as:

Commented [JM8]: Link this to the briefing, I.e. the need to be able to access the internal components to:
-turn systems on or off in case of launch delay
- recharge the battery if necessary
- adjust the transmission frequency (MHz) on the day so it doesn't conflict with other teams.
They will appreciate that you have thought about this.

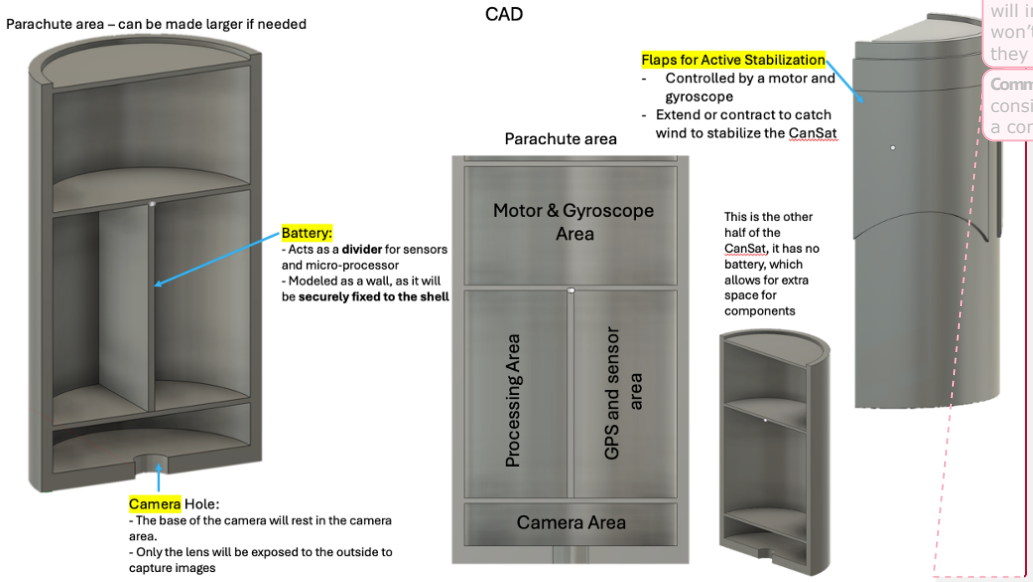
- To turn on or off systems in case of a delay on launch day.
- To recharge the battery if necessary
- To adjust the transmission frequency on launch day to prevent confliction with other teams.
- To simply test if systems are working

For all these reasons, it would be beneficial to us to choose design one, as with design two, all the parts would be locking in place and it would be difficult to mauver them around.

How the space is divided up

Commented [JM9]: Comment on why you have compartmentalised the internal structure, this will improve rigidity and make sure components won't rattle around a lot, reducing damage, as they will experience a lot of forces upon launch.

Commented [JM10]: I think it is important you consider your flap hinge design, even if it is just a comment and not an image of a CAD design

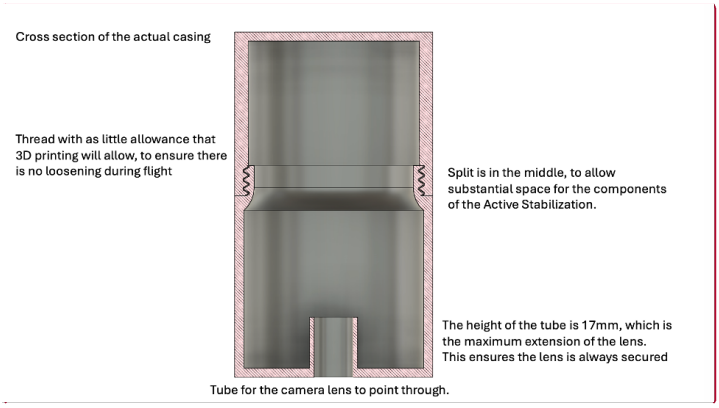


The space is divided up, to improve the structural integrity of the cansat, and to prevent parts from colliding or putting pressure one each other. However, having strict compartments that are pre-printed, would majorly reduce the accessibility of the cansat. Therefore, adding the compartments after the components are in the shell would be a better idea, as it would also be more space efficient, and allow access to all parts, as you simply have to lift a plate to access the component.

Flap Design:



How the actual CAD looks:



Commented [JM11]: Is there any reason why this design doesn't have the compartmentalisation like the top design, is it anything you can comment on as a design choice or a flaw?

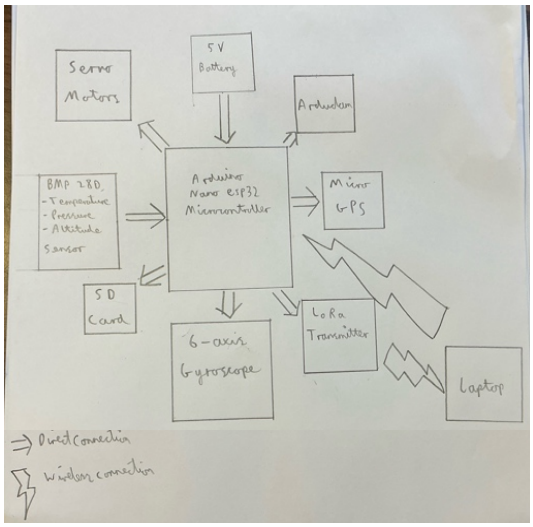
The actual CAD that we are 3D printing for the shell of the CanSat has no compartments, as they would largely reduce the accessibility we would have to the components. Therefore, we have chosen to add the structure supporting pieces after the components.

Materials and justification:

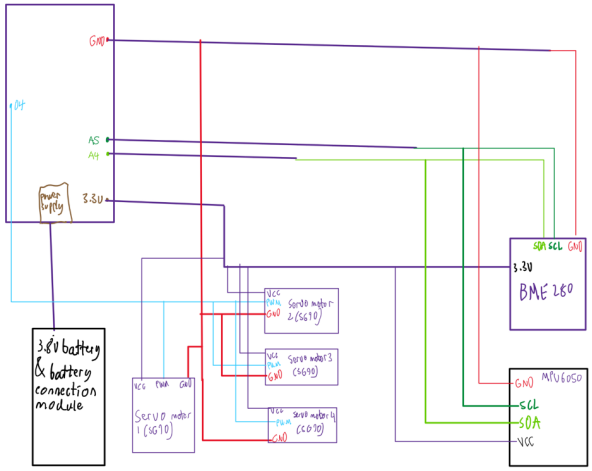
Body – PLA – it is easy to 3D print, therefore easy for us to manufacture and test, whilst also being cheap. Its strength-to-weight ratio can also be improved by using a honeycomb structure which decreases the amount of material we have to use while maintaining its strength.

AS Flaps – Ripstop Nylon – it has a great strength-to-weight ratio, it is elastic so won't break under high drag forces, it is resistant to abrasion, tearing and mildew which is important as the flaps will be under high mechanical stress due to the motors and wind applying forces on it in opposite directions.

3.2 Electrical design



Progress on initial circuit diagram shown above (these are the connections we have done so far):



Arduino Nano – ESP32:

We choose the Arduino nano esp32 due to its compact size which doesn't sacrifice much of the component's performance. We also chose it due to the fact it is a microcontroller. A microcontroller is good for applications which don't require much processing which is not what we require as the processor would only need to either store data in an SD card and relay information to the ground station. However, there will be some processing such as with the servo motors and calculating by how much the servo motors will adjust. We could have chosen the arduino uno R4 or the raspberry pi, but we chose arduino for a first because it is programmed in C++ which is faster than python, more commonly used in the industry of robotics and gives you more control over the program. However, we chose the arduino nano over the arduino uno due to the fact that the arduino uno is a microprocessor, which requires more power and is less efficient than a microcontroller. However, a drawback is that C++ is naturally a more complex language compared to python due to its closer resemblance to machine code. However, we believe it won't be too much of an issue as we are not doing much processing on the microcontroller but simply storing data on an SD card and some calculations on what angle and force the servo motors should be operating.

Commented [JM12]: Can you add headings to break up the sections in the component choices

BMP280

We chose the BMP280 due to its multifunctional capability which will allow us to complete our primary mission much more efficiently while being compact. Its feature of measuring temperature, air pressure and altitude within one component gives us more room for optimising power usage as well as minimising the total mass of the CanSat, aiding our attempts of meeting the criteria set to us at the beginning of this project. Furthermore, the BMP's compatibility to the arduino nano meant it was an obvious choice due to many other options not competing with the C++ language. Alternatively, we could have chosen the BME280 due to its very similar design and functionality, as well as its included measurements of humidity. However, the BME is less accurate than the BMP, resulting in less valid measurements which we wish to use during launch, and we do not require the humidity to be measured for out secondary mission, so the BMP was the better option.

Battery

We have chosen a 3.8v battery however the arduino can only handle 3.3 volts so we will use a resistor to lower the voltage. However, we still require this total voltage for the rest of the CanSat, so researching alternative wouldn't be wise.

Gyroscope

We have chosen a 6-axis gyroscope, as it provides, roll, pitch, and yaw, which is needed for stabilisation. With just the x,y and z coordinates, we wouldn't be able to adjust the angle at which the CanSat is oriented. Therefore, we can use the extra axes that this gyroscope provides us to accurately measure and calculate how much we need to rotate the servos to catch enough wind.
Positives: It is highly accurate, small, and is compatible with the Arduino nano esp32 that we are using. It provides us with the information that we need to perform the stabilisation calculations we need to do.
Negatives: It will be highly complex to code with and is going to cost a significant amount of our budget to buy.
Alternatives: There are a plethora of different components we could have selected, however the compatibility to the arduino nano as well as the additional flexibility in monitoring and controlling the CanSat's movements are why this gyroscope is the best option for our project.

Motor

We have chosen servo motors as they can be programmed to rotate to a specific and accurate angle with a specific force, allowing the fins to be pushed out by the desired amount so the CanSat will catch enough wind to stabilize. This makes servo motors perfect for our use case. This along with the 6-axis gyroscope means we have the full ability to monitor and control the CanSat accurately and effectively.

Transmission

We are using a LoRa as we do not need high speed frequent data transmission. We will be transmitting the location of the cansat from the gps every minute. Due to the close proximity of the CanSat to the ground and our team, as well as the complexity of any components in the satellite being lesser than a high-end satellite, we simply do not require the transmitting power that may be necessary for other devices.

Camera

We will be using the OV6542 camera module as it is compatible with arduino nano esp32(which is quite rare). However, it does not have the best camera quality compared to other cameras. To mitigate this we will aim to take pictures as frequently as possible to be able to produce a high-quality render. This may be quite a drawback to the CanSat's design, however the uncommon compatibility to the arduino nano means that there are little other options to choose from, so mitigations would have to be made. Furthermore, the quality is still able to help us fulfil our secondary mission, so it is not a major issue which we should try to adapt to. If we were to choose a camera with higher resolution, such as the Orion12MP which has a 12-megapixel camera, we would need to account for the higher required storage space and processing power. Our current equipment and components are able to maintain the increased required processing power, and our SD card only has 16 Gigabytes of storage space. This is not enough for the number of images which we need to take for the 3D render. Therefore, it is best for us to stick with the OV6542 camera module and use our plans to mitigate the issues from it.

3.3 Software design

Software required: [Arduino IDE, C++, pixpro, python, pycharm]

We will need a module on the microcontroller in C++ which will receive the temperature, and air pressure data from the BMP280 and will store it on an SD card. This is similar to how we will be collecting the images from

Modules in arduino program needed:

- collect data from BMP180

-collect images from arducam

-receive data from gyroscope

-adjust servo motors

-store in SD card[x]

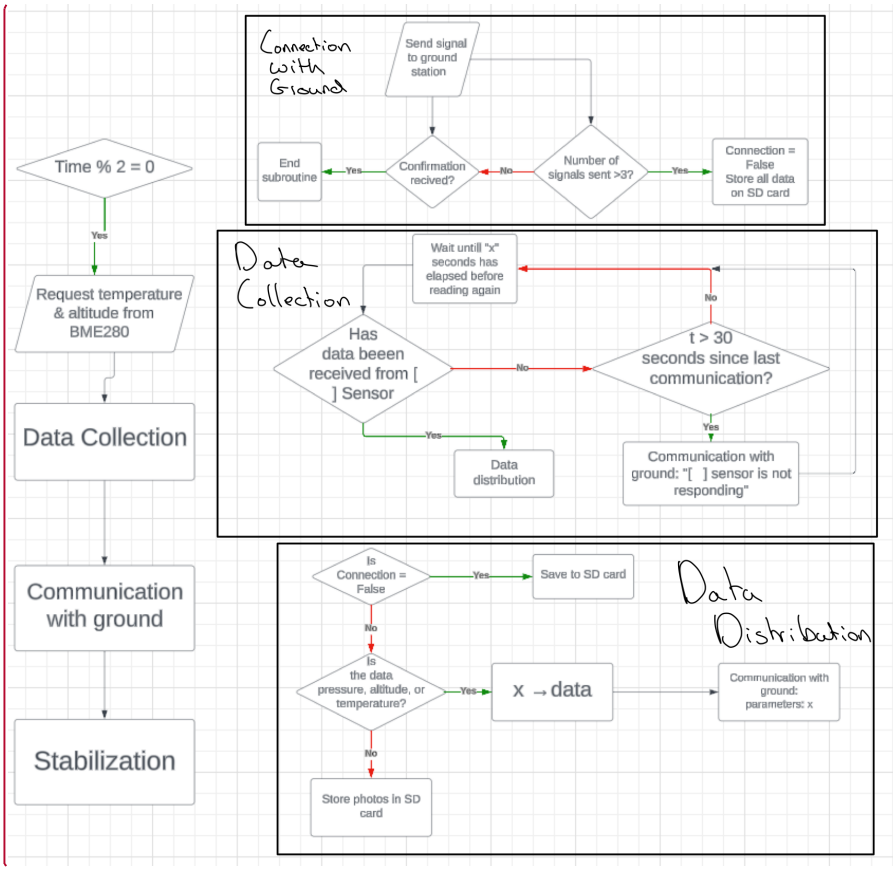
-collect data from GPS

-sending signals with LorA

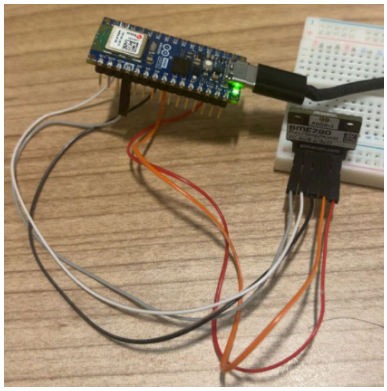
We will be using Matlab to turn the data in the SD cards into a graph.

We will use pixpro to turn the 2D images into a topographical map.

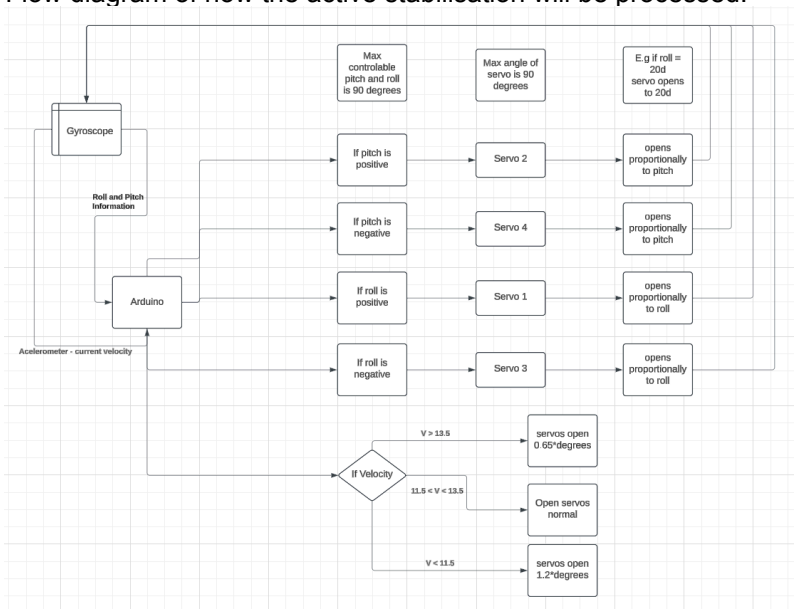
Commented [JM13]: This needs to be made bigger as otherwise it won't be readable



Picture showing Arduino connected to BME 280 which measures temperature and pressure.



Flow diagram of how the active stabilisation will be processed:



3.4 Landing and recovery system

Drag equation:

$$V_{landing} = \sqrt{\frac{2\,Kg}{\rho}} \qquad K = \frac{m}{C_d A}$$

Calculations:

$F_g = mg \qquad F_d = \frac{1}{2} C_w \rho v^2 A$

$mg = \frac{1}{2} C_w \rho v^2 A$

$2mg = C_w \rho v^2 A$

$A = \frac{2mg}{C_w \rho v^2}$

m = mass of CanSat
g = acceleration due to gravitational attraction
C_w = drag coefficient of parachute
ρ = local density ($1.225\ kgm^{-3}$)
A = surface area of parachute
v = decent velocity

$v = \frac{8+11}{2}$

$= 9.5$

mass = 0.35kg

$A_{max} = \frac{2(0.35)(9.81)}{C_w (1.225)(9.5)^2}$

$= \frac{0.062113177768}{C_w}$

$= \frac{0.06211}{C_w}$

$A_{min} = \frac{2(0.30)(9.81)}{C_w (1.225)(9.5)^2}$

$= \frac{0.053239866658}{C_w}$

$= \frac{0.05324}{C_w}$

$= \frac{62160}{1}$

$= 62160 = \pi r^2$

$19770.227 = r^2$

$140 \cdot = r$

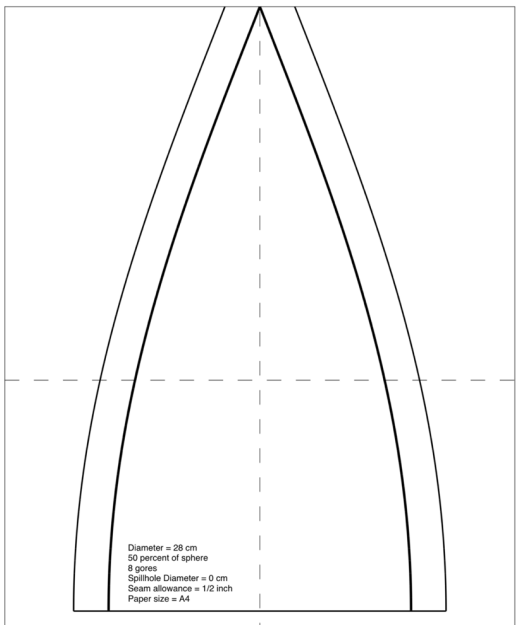
$281.2 = d$

$28.1 km^2$

$= \frac{0.06211}{0.01}$

$m = 0.350$
$A =$
$\rho = (1.225 kgm^{-3})$
$C_w =$
$g = 9.81$
$v = 9.5$

One panel for our parachute template:



We choose to use the a hemisphere design for the parachute as it will provide a good drag coefficient to provide a good decent velocity. This will allow us to take good photos but also be able to land properly.
We have chosen this size by working out the drag coefficient needed for the decent velocity, which was 1. We then used for the formula and find the minimum and maximin area it could be. We then equated this area to the area of a circle to find the diameter. This came out to be around 28cm.
You can find the workings above.

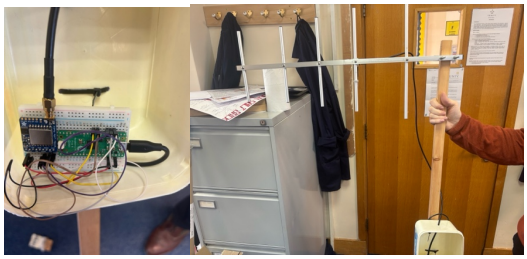
We have chosen the ripstop nylon because it has water resistance and durable and breathable substance.
We chosen to use as it is strong, light weight and durable. These properties are perfect for the rocket as we need it to withstand the force of are resistance and the tension that will be placed on the rope.

Commented [JM14]: You need to put in more details about the parachute design, why have you chosen this template and size, as well as hemispherical shape.
Also comment of materials including the ripstop nylon and dyneema chord to attach parachute to the casing

3.5 Ground support equipment

We will need:

- Laptop – a windows computer that will connect to the Yagi antenna and to store the record the data that is being live transmitter
- Yagi antennas:
    - The Yagi antenna is uni-directional, which allows them to have a much larger range, of up to 5km, with the internal antenna only having to be 17 cm.
    - However, you do have to point the antenna at the cansat while it is in flight, which may prove to be a difficult task, as if we are not aiming it right, we will lose transmission with the cansat and therefore loose some data.



Commented [JM15]: Refer to raspberry pi booklet for more details on how the ground equipment is put together and include the photos you took today of the ground kit from last year.

3.6 Testing

For testing out CanSat, we must put lots of effort into ensuring all components work properly and there are no concerns for the CanSat's performance once we have launch it on our first launch date.

One of our objectives is to have our CanSat match the fundamental rules given for it:

- We will evaluate numerous iterations of layering and organising the components of the satellite to test for the optimum volume space, allowing it to fit within the required volume of 66mm diameter and 115mm length whilst possibly giving excess room for alterations and additions. When experimenting for the optimal formation of components, we will keep in mind that the battery should remain close to the case of the CanSat so an opening can be attached, allowing for easy access to the battery. This will also be done for any power switch we include. Our parachute will be designed to be easily visible from large distances through a neon colour.
- Having any elements included not extend the can's diameter until after launch will be tackled by the optimal space created allowing for or including these elements, then testing for mechanical methods of extending them out of the CanSat for use.
- To ensure our CanSat meets the weight requirements of between 300-350g, the components and elements used should be evaluated on weight as to not cause little room for error when including the case and other features of the satellite.
- We will evaluate the cost of the components which we purchase, trying to fall under the budget of £400, finding a balance between the weight and quality, as well as the cost which comes with it.
- Furthermore, when researching all components that will be used, we will ensure that no potentially hazardous materials are used, maximising safety.
- We are using a 5V battery due to it being optimal for the microcontroller. We can time the duration which the battery lasts every 2-3 weeks (so its not too repetitive and damages the battery) to test if it can last longer than 3 hours, the requirement for the launch day.

Another objective is to test for the overall strength of all components to ensure there are no faults once we get to launch day. We will also test everything before we launch our CanSat, but frequent testing can help recognise errors that we can fix beforehand:

- Parachute: Testing the parachute will involve evaluating its tensile strength, or how much weight it can carry before it snaps and tears. This is important as the parachute must be able to withstand the force which the CanSat will put on it, otherwise the satellite would detach and plummet to the ground. To test for the strength of the parachute, we will evaluate two components of tensile strength: The tensile strength of the rope and the tensile strength of the parachute itself. To test the rope, we will tabulate data for the thickness of the rope against the mass which it can carry before snapping. If it can hold around 5kg, this will approve its ability to hold the required 50N of force under the rules, as well as comfortably put us in the strength required to hold the weight of the CanSat; choosing the optimal thickness of rope will decrease the mass of it, reducing the total mass of the satellite. To ensure nothing goes wrong, we will test the parachute multiple times for prolonged length of time to ensure it will not fail during initiation.
- In order to make the CanSat descend between 10-15m/s, we should also test the parachute for different sizes and tabulate how its surface area corresponds to its air resistance. This is key as a larger surface area would result in more air resistance (and less for a smaller one), so we must somewhat limit the size of the parachute to allow it to still hold the required force but not exceed the maximum velocity of descent by being too small.
- When the rocket which takes the CanSat up launches, the satellite will experience an acceleration of up to 20g, meaning it should consist of materials tough enough to withstand the force. This could prove to be a challenge as tougher materials are often denser, resulting in the satellite potentially exceeding 350 grams. We can research on materials which have a high strength-to-weight ratio, allowing us to make a stronger satellite. Furthermore, finding the right process for making the case will impact its strength against the 20g force; for example, if metal was chosen, welding could be more effective than using nuts and bolts as less tensile stress if therefore put on the pieces.
- Furthermore, the parachute, although not required to be in the volume of the can, should be protected within a case as to not tear off during launch. We can do this effectively by having a small opening at the top of the CanSat where the parachute will go. The air resistance and force of gravity will prevent the parachute from flying out of the capsule; once the satellite reaches the needed altitude, the decrease velocity of the CanSat will decrease the air resistance, allowing the parachute to exit the capsule. To prevent the satellite from tipping over and stopping the parachute from opening properly, we can concentrate the mass of the components to the bottom of the satellite, resulting in it remaining upright due to the centre of mass being below the centre of volume.

A third objective focuses on launch day, as the processes taken must go almost flawlessly as to allow us to proceed to the next stage of the competition. This includes making sure the satellite doesn't break, having all function active and operable when in the air, etc:

- We must ensure the CanSat is either easy to see through a flickering light or has a tracking device to lower the chance of loosing the satellite
- We will allow the CanSat to alter its frequency (within UK regulations) to allow for ensured communications with the ground, as well as our ability to use the functions of the satellite.

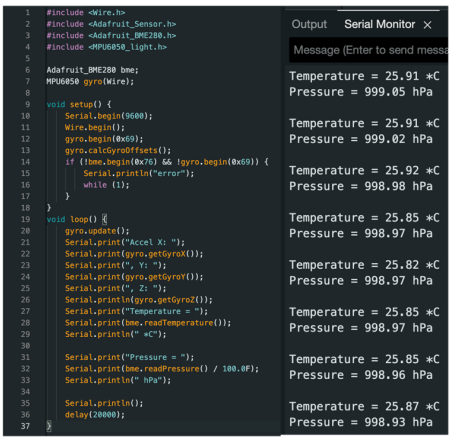
Component Testing Plan:

Things to test:

Commented [AG16]: Can you provide the methodology you will use for each test, have you collected any results that you can give?

- Casing durability
    - drop from a second story height with no parachute.
- Gyroscope response
    - Test code by angling gyroscope at known angels and validating results. Repeat this for roll and pitch, for a minimum of 3 angels per axis.
- Servo response
    - Test if servos react to change in roll and pitch. Check if the correct servos are responding
- Camera taking photos(OV6542)
    - set the camera to take photos for 2 minutes, then open the SD drive on a computer, and check if the photos are correct
- BME 280 response:
    - Run the BME280 code and sense check the results to what room temperature should be and what the atmospheric pressure is
- Parachute testing:
    - Throw the cansat with the parachute attached off a high point. Repeat this multiple times and determine whether the parachute is effectively slowing the cansat down.
- Transmitter
    - Stand up to 600m away, as that is the furthest the cansat could go
    - Turn on the system and record some temperature and pressure readings.
- Laptop
    - Ensure the laptop has the Arduino IDE installed to receive the sensor data
- SD card
    - Take known pictures, and record known data a certain number of times, and verify if that information is present on the SD card
- Servo motors
    - Ensure they can hold the string around themselves, and release a specific desired amount every time the roll or pitch changes
- GPS
    - Go to a location you know the coordinates and altitude of, turn on the gps, take some readings and verify if they are correct

Testing Results:



```
#include <Wire.h>
#include <Adafruit_Sensor.h>
#include <Adafruit_BME280.h>
#include <MPU6050_light.h>

Adafruit_BME280 bme;
MPU6050 gyro(Wire);

void setup() {
    Serial.begin(9600);
    Wire.begin();
    gyro.begin(0x69);
    gyro.calcGyroOffsets();
    if (!bme.begin(0x76) && !gyro.begin(0x69)) {
        Serial.println("error");
        while (1);
    }
}
void loop() {
    gyro.update();
    Serial.print("Accel X: ");
    Serial.print(gyro.getGyroX());
    Serial.print(", Y: ");
    Serial.print(gyro.getGyroY());
    Serial.print(", Z: ");
    Serial.println(gyro.getGyroZ());
    Serial.print("Temperature = ");
    Serial.print(bme.readTemperature());
    Serial.println(" *C");

    Serial.print("Pressure = ");
    Serial.print(bme.readPressure() / 100.0F);
    Serial.println(" hPa");

    Serial.println();
    delay(20000);
}
```

```
Output  Serial Monitor ×
Message (Enter to send messa
Temperature = 25.91 *C
Pressure = 999.05 hPa

Temperature = 25.91 *C
Pressure = 999.02 hPa

Temperature = 25.92 *C
Pressure = 998.98 hPa

Temperature = 25.85 *C
Pressure = 998.97 hPa

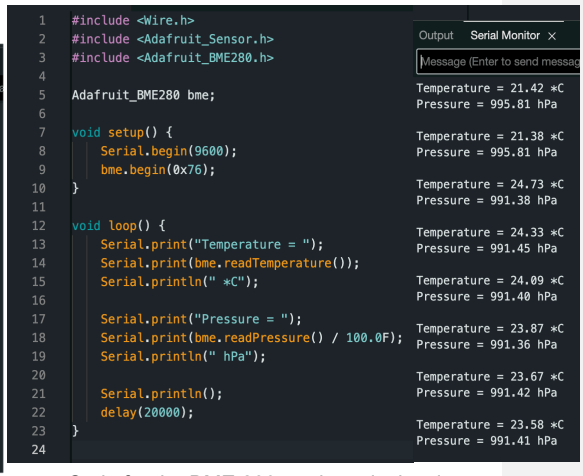
Temperature = 25.82 *C
Pressure = 998.97 hPa

Temperature = 25.85 *C
Pressure = 998.97 hPa

Temperature = 25.85 *C
Pressure = 998.96 hPa

Temperature = 25.87 *C
Pressure = 998.93 hPa
```

This is updated code showing the accelerometer with an output that has not changed, as there is an error in the wiring.



```
#include <Wire.h>
#include <Adafruit_Sensor.h>
#include <Adafruit_BME280.h>

Adafruit_BME280 bme;

void setup() {
    Serial.begin(9600);
    bme.begin(0x76);
}

void loop() {
    Serial.print("Temperature = ");
    Serial.print(bme.readTemperature());
    Serial.println(" *C");

    Serial.print("Pressure = ");
    Serial.print(bme.readPressure() / 100.0F);
    Serial.println(" hPa");

    Serial.println();
    delay(20000);
}
```

```
Output  Serial Monitor ×
Message (Enter to send messag
Temperature = 21.42 *C
Pressure = 995.81 hPa

Temperature = 21.38 *C
Pressure = 995.81 hPa

Temperature = 24.73 *C
Pressure = 991.38 hPa

Temperature = 24.33 *C
Pressure = 991.45 hPa

Temperature = 24.09 *C
Pressure = 991.40 hPa

Temperature = 23.87 *C
Pressure = 991.36 hPa

Temperature = 23.67 *C
Pressure = 991.42 hPa

Temperature = 23.58 *C
Pressure = 991.41 hPa
```

Code for the BME 280, and result showing temperature and pressure.



Parachute Testing:
We threw the cansat as hard as we could with an upwards trajectory, then analysed the stages the parachute goes through.

Each frame is described as follows, going left to right:

1. CanSat is launched upwards
2. Parachute starts to open
3. Cansat reaches its peak
4. Cansat swings violently
5. Cansat settles but still has still in pitch and roll

Subroutine for active stabilisation:

Commented [AG17]: Can you annotate this or make it simpler so someone like me with limited programming knowledge knows what is going on?



```
#include <Adafruit_BME280.h>
#include <Wire.h>
#include <MPU6050.h>
#include <Math.h>
//#include <Servo.h>

Adafruit_BME280 bme;
MPU6050 mpu;
int16_t ax, ay,az, gx, gy, gz;
Servo motor1;
Servo motor2;
Servo motor3;
Servo motor4;
int motorAngle1;

void setup() {
  myservo.attach(4);
  Serial.begin(9600);
  Wire.begin();
  mpu.initialize();
mpu.setSleepEnabled(false); // Ensure wake-up mode
mpu.setFullScaleAccelRange(MPU6050_ACCEL_FS_2); // ±2g range
mpu.setFullScaleGyroRange(MPU6050_GYRO_FS_250); // ±250°/s range
mpu.CalibrateAccel(6);  // 6-point calibration
mpu.CalibrateGyro(6);
```

1

```
  // Verify connection
  Serial.print("MPU6050 Test Connection: ");
  Serial.println(mpu.testConnection() ? "OK" : "FAILED");
  // Initialize BME280
  if (!bme.begin(0x76)) {
  Serial.println("BME280 not found!");
  while(1);
  } else {
  bme.setSampling(Adafruit_BME280::MODE_NORMAL,
                  Adafruit_BME280::SAMPLING_X2,  // Temp
                  Adafruit_BME280::SAMPLING_X16, // Pressure
                  Adafruit_BME280::SAMPLING_X1,  // Humidity
                  Adafruit_BME280::FILTER_OFF,
                  Adafruit_BME280::STANDBY_MS_1000);
  Serial.println("BME280 Configured!");
  }

  }

  void loop() {

  // reading temperature and pressure
  float temp = bme.readTemperature();
  float pressure = bme.readPressure();
  Serial.print("Temp(*C):");
  Serial.println(temp);
  Serial.print("Pressure(Pa):");
  Serial.println(pressure / 100);
```

2

```
  mpu.getMotion6(&ax, &ay, &az, &gx, &gy, &gz);

  Serial.print("Accel (X, Y): ");
  Serial.print(ax/ 16384); Serial.print(", ");
  Serial.print(ay / 16384); Serial.print(", ");

  Serial.print("Gyro (X, Y): ");
  Serial.print(gx / 131); Serial.print(", ");
  Serial.print(gy / 131); Serial.print(", ");

  delay(2000);

}

void stabilisation(int motor1, int motor2, int motor3, int motor4, gx, gy, ax, ay, a
{
  unsigned long currentTime = micros();
  float dt = (currentTime - lastTime) / 1000000.0;  // in seconds
  lastTime = currentTime;

  // Calculate pitch and roll from accelerometer
  float accelPitch = atan2(-ax, sqrt(ay * ay + az * az)) * 180.0 / M_PI;
  float accelRoll = atan2(ay, sqrt(ax * ax + az * az)) * 180.0 / M_PI;
```

3

```
// Integrate gyroscope data to update pitch and roll
pitch = gy * dt;  // gyroY = pitch rate
roll  = gx * dt;   // gyroX = roll rate

// Complementary filter to combine accel and gyro data
float alpha = 0.98;  // Trust gyro 98%, accel 2%
pitch = alpha * pitch + (1 - alpha) * accelPitch;
roll = alpha * roll + (1 - alpha) * accelRoll;
if (pitch < 0){
  motor1.write(-pitch);
}
else if (pitch > 0) {
  motor2.write(-pitch);
} else if (roll < 0){
  motor3.write(-roll);
} else if (roll > 0) {
  motor4.write(-roll);
```

4

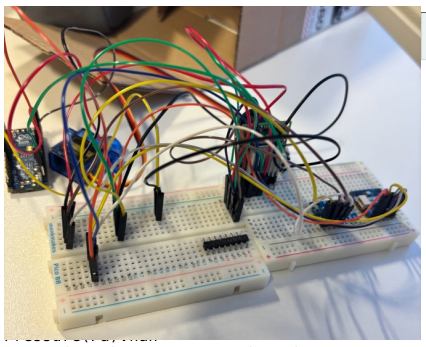
Picture 1:
We are importing libraries (pre written code with functions that will help us to make our code more efficient when communicating with sensors) to use later in the program. We then declare our variables and functions to use later in the program. We then connect with and calibrate the gyroscope in the setup function which is a block of code which only runs once when the code starts running.

Picture 2:
(Still in the setup function) We then test the connection of our sensors, after which we connect with and calibrate the BME280.
Now we are in the loop function which runs continuously in a loop forever (or until a new program is loaded onto the arduino). We use the prebuilt code from the libraries to take readings of the temperature and air pressure from the BME 280 and store it in variables to output. We then get the acceleration and angular velocity in 3 dimensions. We then divide the values by the respective acceleration and angular velocity due to gravity and output it.

Picture 3 & 4:
We are now outside the loop function and have created a function to stabilise the CanSat. We calculate the pitch and roll from the accelerometer and then find the rate of change in angular velocity. We use a complimentary filter alpha because the angular velocity readings are usually more accurate over short time periods whereas the accelerometer is less accurate, so we have balanced the weighting of 98% accelerometer pitch and 2% angular velocity pitch. We then add the weighted pitches together to get the actual pitch we will be using for the stabilisation. We then repeat with the roll. After calculating the pitch, we check if the pitch and roll are positive and negative respectively, for the pitch rightwards is positive and leftwards is negative, for the roll tilting frontwards is positive and backwards is negative. 2 servo motors are responsible for the pitch and 2 more are responsible for the roll. The code is not yet completed so we are yet to assign which motor is responsible or their respective pitches, e.g. motor 1 is on the right side of the CanSat so if the pitch is positive it will move with the pitch to open the flap. We will test this by using a hair dryer at different speeds to see if it can maintain an accurate and consistent reading.

Testing BME280 and MPU6050 for temperature, air pressure, pitch and roll:

Our BME 280 gave no readings after we rotated the circuit to test the gyroscope, because of this we will investigate it's suitability for this project by taking readings at different altitudes, speed and rotation. We will also further investigate the gyroscopic data by rotating the CanSat quickly and testing the gyroscope under different speeds and altitudes



Arduino Nano ESP32
Port: /dev/cu.usbmodem744DBDA0A8A42

```
.MPU6050 Test Connection: OK
BME280 Configured!
Temp(*C):23.29
Pressure(Pa):1015.00
Accel (X, Y): 0, 0, Gyro (X, Y): 0, 0, Temp(*C):23.30
Pressure(Pa):1015.02
Accel (X, Y): 0, 0, Gyro (X, Y): 0, 0, Temp(*C):23.28
Pressure(Pa):1015.01
Accel (X, Y): 0, 0, Gyro (X, Y): 0, 0, Temp(*C):23.28
Pressure(Pa):1014.98
Accel (X, Y): 0, 0, Gyro (X, Y): 0, 0, Temp(*C):23.28
Pressure(Pa):1015.01
Accel (X, Y): 0, 0, Gyro (X, Y): 0, 0, Temp(*C):23.28
Pressure(Pa):1015.04
Accel (X, Y): 0, 0, Gyro (X, Y): 0, 0, Temp(*C):23.27
Pressure(Pa):1015.03
Accel (X, Y): 0, 0, Gyro (X, Y): 0, 0, Temp(*C):23.27
Pressure(Pa):1015.02
Accel (X, Y): 0, 0, Gyro (X, Y): 0, 0, Temp(*C):23.27
Pressure(Pa):1015.03
Accel (X, Y): 0, 0, Gyro (X, Y): -1, 2, Temp(*C):nan
Pressure(Pa):nan
Accel (X, Y): 0, 0, Gyro (X, Y): -17, 64, Temp(*C):nan
Pressure(Pa):nan
Accel (X, Y): 0, 0, Gyro (X, Y): -5, 12, Temp(*C):nan
Pressure(Pa):nan
Accel (X, Y): 0, 0, Gyro (X, Y): -2, 1, Temp(*C):nan
Pressure(Pa):nan
Accel (X, Y): 0, 0, Gyro (X, Y): 1, 1, Temp(*C):nan
Pressure(Pa):nan
Accel (X, Y): 0, 0, Gyro (X, Y): 2, 2, Temp(*C):nan
```

The code is not yet completed so we are still to program the camera, SD card module and GPS module

3.7 Overall testing for launch

- We will test the LoRa module over different distances to see maximum distance we can communicate with the antenna and the LoRa module
-We will leave the battery on until it dies to see how long the battery can last
-we will test the parachute at different heights on a replica cansat of same weight and dimensions to see what height the parachute should be released at
-we will test the camera module at different heights to see the quality of images
-we will test the GPS by putting the CANSAT in random places and seeing if the location it sends via the LoRa is accurate
-we will test the BMP280 by putting it in different temperatures, air pressures and altitudes and seeing if it gives the correct measurements.
-we will test the servo motors by seeing if they change angle upon signal by the arduino nano esp32 and we will see if they make a difference when we test the parachute with the cansat

3.8 Evidence of CanSat build*






This shows the CanSat shell and the parachute. As shown in the cad, there is a thread on the bottom and top, which splits the can in half, allowing easy access to both halves. The bottom also has the extended hollow cylinder that will hold the camera module in place. To connect the parachute to the body, we used 8 eye screws, temporarily tying the string to the loops for testing, but using a more permanent solution like sanding the eye and supergluing the knot to the eye. The screw is smooth but still is secures the two halves together tightly once past a certain point. There is also a spill hole at the apex of the parachute which prevents the entire system from catching too much wind and drifting too far from the launch site.

4 Outreach programme

In order to promote our idea to the masses, we have entered the realm of social media. We have created a TikTok account and an Instagram account in order to gain awareness for our project and our concept. On these accounts we will post videos and photos related to our project and our work to gain a following to present our ideas to the public.

We plan on using modern comedy techniques to gain traction and gain a following for engineering. We have also given a multi-week educational talk to the next generation of engineers at our school's Young Engineer's Society to spark interest and instil new knowledge upon them. This included giving them an introduction into what CanSat is, and how it mimics the real world. We then gave them a number of secondary missions to choose from, with an accompanying brief to make it easier for them to conceptualise designs. After designing their CanSat's secondary mission we provided them with a limited supply of typical arts and crafts materials to make a prototype to prove their design concept. They received sufficient time to reach their final product, during which, we walked around assisting the groups, providing them with some small pieces of information that help them develop their engineering abilities by uncovering new techniques, for example using paper or card folded over each other to form a spring which was necessary for one of the briefs. After making their models, they presented their models, to the rest of the society, with the top 3 groups winning a bag of sweets.
We plan to give another talk at our schools Computer Science society in the near future, which will delve deeper into the programming, and technical aspects of the CanSat which we were unable to cover with the younger students.



We also made a flyer and distributed out to get people interested into the project. We hung pictures up in busy areas and we saw some people being interested into the flyers.
We also posted this on our social medias and gained some good traction in the form of reposts, likes, shares, and comments.
We had many people saying they would like to learn more about CanSat, and multiple year 7,8 and 9s come up to us and ask questions specifically about how we attempt to achieve our secondary mission

5 LAUNCH DAY PREPARATION*

5.1 Launch checklist/countdown

Task	Time	Duration	Team Member responsible
Unscrew cansat, and ensure all the parts are in the correct locations	12:30	10 minutes	Student 2
Plug in Arduino to laptop to check programming is running as expected	12:45	15 minutes	Student 1
make sure battery is fully charged.	13:00	1 minutes	Student 3
turn on power supply (battery and check all components are working)	13:01	4 minutes	Student 4
adjust transmission frequency so that it doesn't conflict with other teams	13:05	5 minutes	Student 5
calibrate and check any sensors	13:10	6 minutes	Student 1
quick check that photos are being taken and stored to SD card	13:16	7 minutes	Student 6
seal casing and check integrity	13:23	2 minutes	Student 2
final check of data transmission on ground station before handing over to rocket team.	13:25	5 minutes	Student 4

5.2 Post mission checklist

Task	Time	Duration	Team member responsible

retrieve satellite	13:35	5 minutes	2. gets cansat
check structural integrity	13:40	5 minutes	2. checks structural integrity
Disconnect parachute	13:47	7 minutes	3. disconnects it 5. Takes the main body
Open casing and turn off power supply	13:49	2 minutes	5. opens the cansat. 6. disconnects the battery
check Li-ion battery is undamaged	13:50	1 minute	6. checks for damage
retrieve SD card and download to laptop	14:00	10 minutes	1. Receives cansat and removes SD card and connects it to laptop
collect data from ground station and save files with photos taken	14: 15	15 minutes	2&3. Go to ground station and saves information

5.3 Launch Day risk log

(Think about what could go wrong on launch day and how the risks could be mitigated)

5.3 Results analysis procedure

Description of procedure of how you are to interpret and use your sensor data for use in your presentation. Include details of any calculations used and how this analysis relates to your primary/secondary mission objectives. How will you determine if your measured data is valid and what is your criteria for a successful mission?

6 Lessons learned*

Reflect on your participation in the competition. What have you learnt from the process? Has this changed anything for you in terms of subject or career decisions? What has been the most challenging part? What has been the best part?

*Only required for final report